Challenge 4: Sidorenko Conjecture for Half-Graphs

Definition 1 (Graph homomorphism density)**.** Let G and H be simple graphs.

1. A function $f : V(H) \to V(G)$ is a *graph homomorphism* from H to G if for each edge $uv \in E(H)$, we have $f(u)f(v) \in E(G)$.

2. Let $\hom(H, G)$ be the number of graph homomorphisms $H \to G$.

3. The *homomorphism density* of H in G is the fraction

$$t_H(G) := \frac{\hom(H, G)}{|V(G)|^{|V(H)|}}$$

Conjecture 1.1 (Sidorenko's Conjecture [1])**.** For every bipartite graph H and for any graph G, we have:

$$t_H(G) \geq t_{K_2}(G)^{|E(H)|}$$

We are interested in the special case of the conjecture, where $H = H_r$ is a half-graph with $2r$ vertices, as follows. As far as we are aware, the conjecture has not been asked anywhere for half-graphs, and the only reason we do it for half-graphs here is that we believe that they are somewhat generic, and it will help us formulate an easier scalable version.

Definition 2 (Half-graph)**.** Let $A_r = \{a_1, \ldots, a_r\}$ and $B_r = \{b_1, \ldots, b_r\}$ be disjoint sets of size r. A *half-graph* is the bipartite graph on the vertex set $A_r \cup B_r$ with the following edges present: $E = \{a_i b_j : i \leq j\}$ (Note that these are about 'half' of the possible edges between A_r and B_r, hence the name).

Conjecture 1.2 (Scaled weakening of Sidorenko's Conjecture)**.** Let $r \in \mathbb{N}$. For any graph G, we have:

$$t_{H_r}(G) \geq t_{K_2}(G)^{(r^2+r)/2}$$

References

[1] A. F. Sidorenko. "A Correlation Inequality for Bipartite Graphs". In: *Graphs and Combinatorics* 9.2–4 (1993), pp. 201–204. DOI: `10.1007/BF02988307`. URL: `https://doi.org/10.1007/BF02988307`.